

Wireshark: Tutorial on Using the Wireshark Tool

Assignment study PDF — English

1. What is Wireshark?

Wireshark is a network protocol analyzer used to capture and inspect packets. It is useful for troubleshooting, learning protocols, monitoring traffic and investigating network behavior.

2. Capture workflow

Open Wireshark, select the appropriate network interface, start capture, generate traffic, stop capture, and then inspect packets. Saving captures as PCAP/PCAPNG files allows later analysis.

3. Packet structure

Wireshark presents captured traffic through packet-list, packet-details and packet-bytes views. Selecting a packet reveals protocol layers and field values.

4. Display filters

Useful examples include `ip.addr == 192.168.1.10`, `tcp.port == 80`, `dns`, `http`, `icmp`, and `tcp`. Display filters change what is shown without deleting captured packets.

5. Follow a conversation

For supported protocols, right-click a packet and use the Follow feature to inspect a conversation or stream. This is useful for understanding client-server exchanges.

6. Practical analysis

A basic exercise is to capture a ping, identify ICMP packets, inspect source and destination addresses, then compare request and reply packets.

7. Safety and privacy

Packet captures can contain sensitive information. Capture only traffic you are authorized to monitor and protect PCAP files appropriately.